Steady Demo Login, Synthetic Population and Deterministic Planning Engine

Code handoff for role-selectable demo access, 240 fabricated longitudinal patient profiles, regional and demographic analytics, planning rules, governance, APIs, screens, tests and release controls.

Scope	Decision
Demo roles	Patient, clinician, reviewer, payer, organization and demo admin
Population	240 fabricated adults, 60 per U.S. Census region
History	Six seeded months of measures, check-ins, modules, gates, actions and outcomes
Learning posture	Descriptive and deterministic planning first; person-level treatment remains clinician-led
Data boundary	No PHI, no copied records, no public enrollment, no production credentials

Build the demo so every number can be traced to a fabricated event, every role sees only its projection, and every planning signal can be inspected, challenged and turned off.

Prepared for product, design, application engineering, data engineering, analytics, security, clinical review and investor demonstrations.

BUILD DIRECTION

Executive decision

This work adds a believable demonstration layer without weakening Steady's permanent architecture. The login screen selects a demo role. The server creates a role-bound session. Every page reads a projection built from the same fabricated event ledger. A nightly reset returns the environment to a known state.

The population-learning layer answers four questions: who uses Steady, how usage differs by region and demographic group, which modules are associated with better observed results, and where a controlled pilot should test a change. It does not decide what treatment a person should receive.

The first release uses deterministic metric definitions and versioned signal rules. Statistical models can run later in shadow mode, but they cannot alter safety gates, treatment access, clinician judgment or patient routing.

The four regions are the U.S. Census regions: Northeast, Midwest, South and West. Each receives exactly 60 profiles. Organization is included as a demo role because organization dashboards already exist in the web design.

ADMIN SCOPE

Demo admin can inspect every fabricated tenant, person, event, projection and test control. Production administration must use purpose-limited permissions and break-glass access. Do not carry the demo admin's blanket visibility into production.

Definition of done

- A presenter can choose any role and reach its correct landing page in two actions.
- Every dashboard contains populated, coherent charts with visible definitions and fabricated-data labeling.
- Every patient has a plausible six-month story that stays internally consistent across patient, clinician, organization and payer views.
- Every aggregate can be rebuilt from the event ledger and matches the live projection byte for byte.
- No planning signal can become a patient-level care order.

SAFETY AND TRUTH

Non-negotiable boundaries

Boundary	Required behavior	Prohibited behavior
Data origin	Generate all people, events and free text from fixed seeds	Copy, perturb or resemble any real patient record
Demo labeling	Show fabricated-data banner on login, shell, exports and charts	Let screenshots look like production patient data
Safety	Use the same deterministic gate engine and most-restrictive-rule precedence	Create a separate relaxed demo safety path
AI	Label summaries, suggestions and hypotheses; require human review	Diagnose, prescribe, clear gates or claim causality
Demographics	Use for representation, disparity audit and aggregate planning	Use race or protected status to restrict or select person-level care
Outcomes	Describe observed change with definitions, windows and missingness	Call one session or one correlation proof of treatment effect
Admin	Allow broad visibility only inside isolated demo tenancy	Implement production superuser access from the demo shortcut
Reset	Rebuild from manifest and seed; write an audit event	Hand-edit demo rows until the story cannot be reproduced

The simulated BLS Part 6 path remains visibly simulated. Steady continues to support preparation and stabilization. Target selection, trauma reprocessing, diagnosis, treatment formulation and final care decisions remain clinician-led.

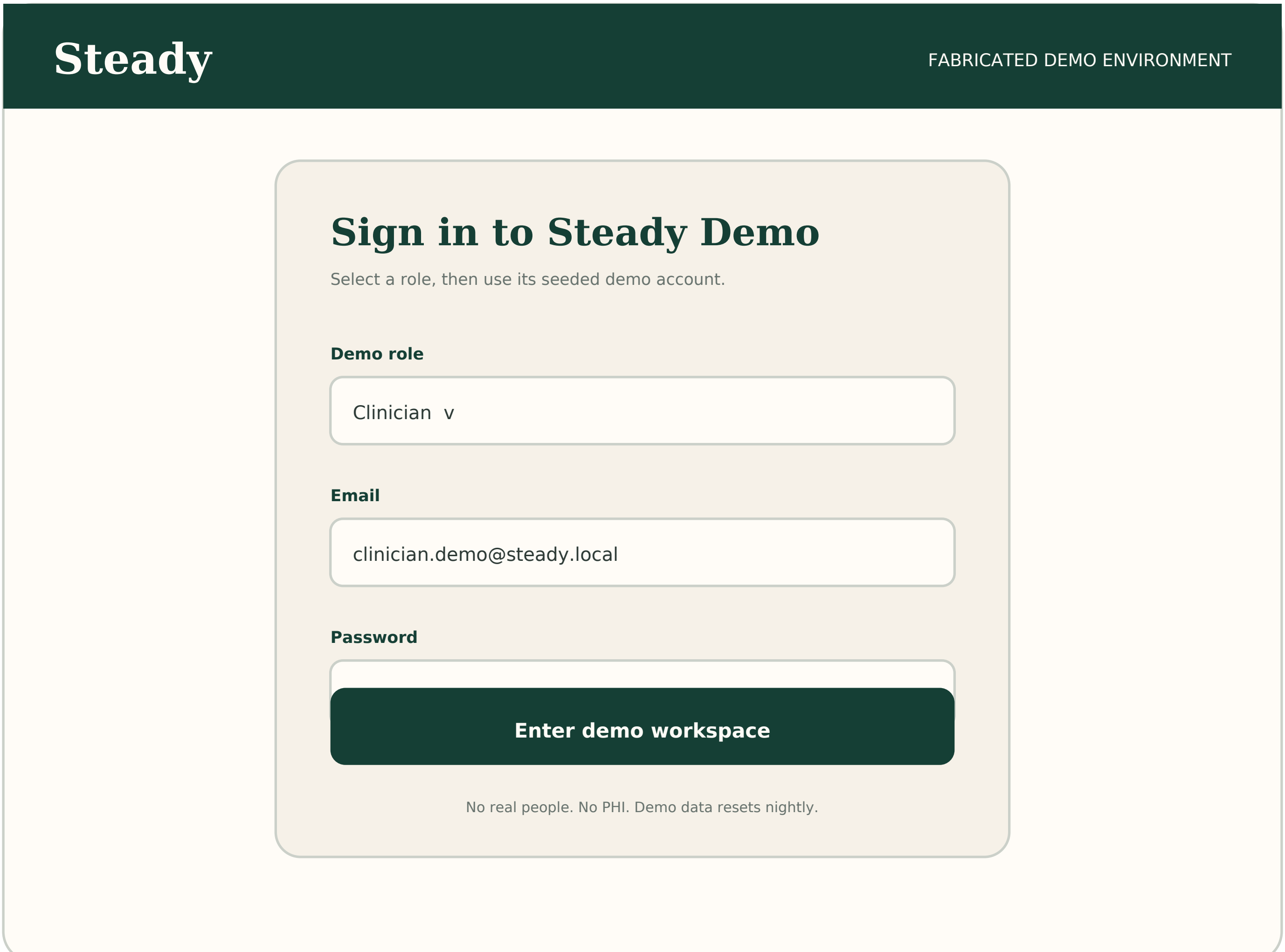
SECTION 1

Role-selectable demo login

A presenter chooses a role, authenticates with a seeded account and enters a role-bound projection. The dropdown is a demo convenience, not the authorization source.

DESIGN RULE	**Action first. Meaning second. Evidence third. Raw records last.**

1.1 Login screen specification



Element	Behavior	Acceptance
Role dropdown	Patient, Clinician, Reviewer, Payer, Organization, Demo Admin	Keyboard accessible; selection never grants permission
Email	Role-specific demo alias	Server validates account-role match
Password	Secret loaded from environment or demo identity provider	Never commit a password to source or client bundle
Demo helper	Copies the selected alias and focuses password	Disabled outside demo environment
Banner	Fabricated demo environment	Persists on every authenticated screen
Submit	Exchanges credentials for a short-lived role session	Server sets tenant, subject, role, purpose and allowed projections

1.2 Demo roles and landing pages

Role	Default account	Landing route	What the role can see	What the role cannot see
Patient	patient.demo@steady.local	/app/today	Self, own actions, own measures, own evidence	Other people, aggregate comparisons, model internals
Clinician	clinician.demo@steady.local	/clinician/today	Assigned panel, safety queue, cited summaries, actions	Unassigned patients, payer cost model, system config
Reviewer	reviewer.demo@steady.local	/review/safety	Fixed gates, evidence, replay, corrections, audit	Routine treatment decisions, credential management
Payer	payer.demo@steady.local	/payer/overview	Aggregate access, engagement, outcomes, utilization, cost	Patient-level clinical records or person search
Organization	org.demo@steady.local	/organization/overview	Aggregate access, capacity, outcomes, service gaps	Payer-wide data or unrelated organizations
Demo Admin	admin.demo@steady.local	/admin/demo	All fabricated tenants, roles, people, events, reset and QA	Any production environment or real data

The role selector provides context and reduces demo friction. Authorization still comes from the server-side identity, tenant and permission claims. A user cannot select Demo Admin and authenticate with the clinician account.

1.3 Demo account and session contract

Recommended account design

- Create one identity per role and optional presenter identities per audience. Do not use one account with a mutable role claim.
- Store passwords or identity-provider bootstrap secrets outside source control. Rotate before each external review cycle.
- Issue sessions with environment=demo, dataset_version, tenant_id, person_id when relevant, role, purpose, issued_at, expires_at and allowed_projection list.
- Expire sessions after 60 minutes of inactivity and 8 hours absolute. Clear all session state when the presenter switches roles.
- Disable password reset, public signup and cross-environment federation in the demo tenant.

Session claim example

```
{ "sub": "demo-clinician-01", "environment": "demo", "dataset_version": "demo-population-v1", "tenant_id": "demo-org-west-01",
"role": "clinician", "purpose": "investor_demo", "allowed_projections": ["clinician_queue.v3", "person_summary.v4"], "expires_at":
"2026-08-29T20:00:00Z" }
```

The client may display the selected role. It must never calculate authorization from the dropdown, URL, hidden field or local storage.

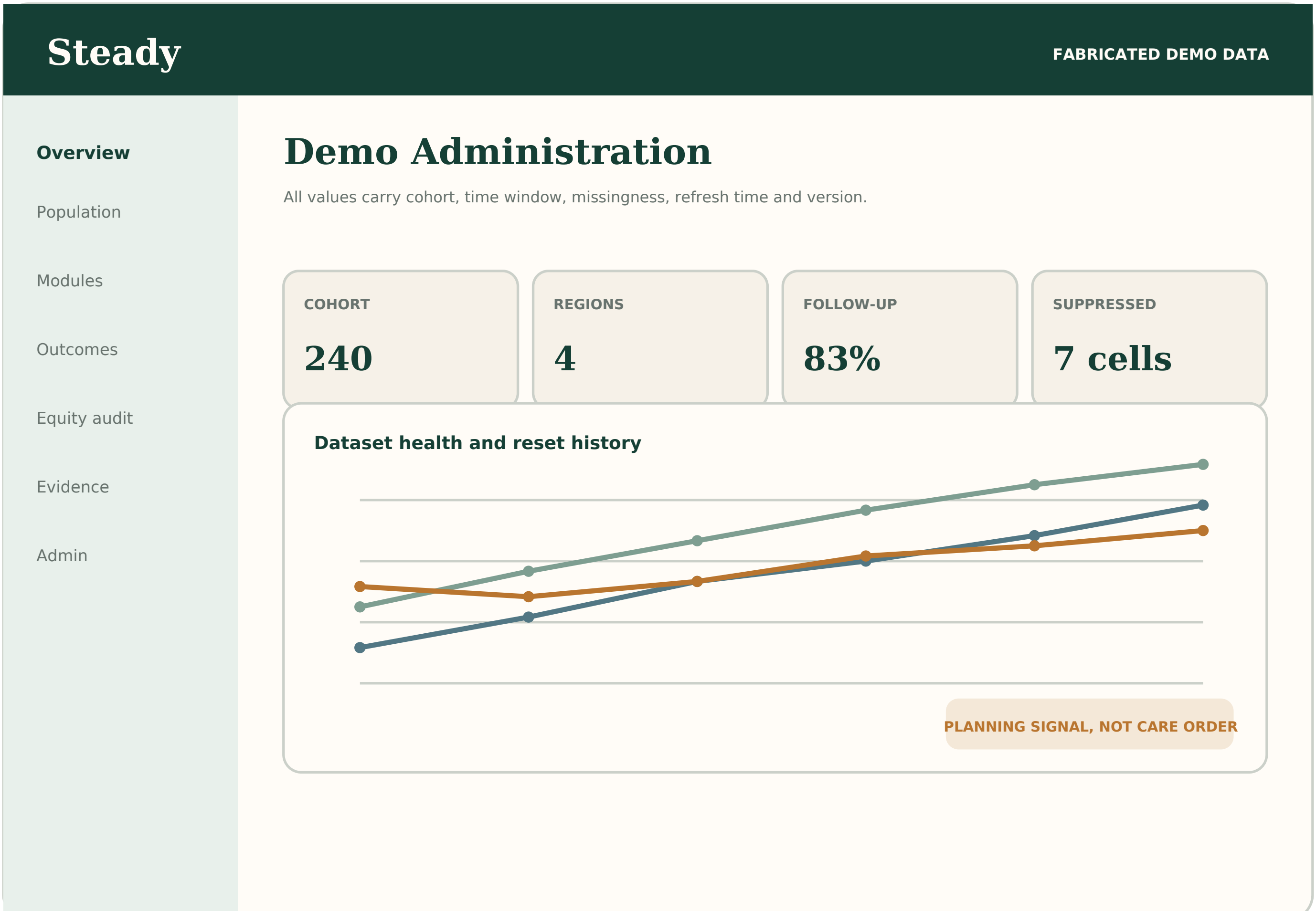
1.4 Authentication and role-switch sequence

Step	Client	Server	Audit	
1	Load demo roles from public configuration	Return enabled demo personas only	Record configuration version	
2	Select role and enter credentials	Authenticate exact identity	Record success or generic failure	
3	Submit requested demo purpose	Resolve tenant, role and allowed projections	Record session issuance	
4	Open landing route	Return role-safe shell and dataset label	Record route and projection version	
5	Switch role	Destroy local state and reauthenticate	Revoke old session before new issue	
6	Reset demo	Admin requests reset with reason	Rebuild data and projections from seed	Record reset

Required negative tests

- Clinician credentials plus Admin selection returns the same generic login failure as any other invalid pairing.
- Payer session cannot request person_summary, even by direct API call.
- Patient session cannot change person_id in the URL and discover whether another subject exists.
- Role switch cannot retain cached patient data from the prior role.
- Demo accounts cannot authenticate against production endpoints.

1.5 Demo admin control center



Control	Behavior	Guard
Reset dataset	Recreate all seed records and projections	Typed confirmation plus reason
Advance clock	Move demo date to a scripted milestone	Demo only; clock shown in shell
Inject scenario	Apply approved event bundle such as a safety pause	Bundle is versioned and reversible by reset
Validate projections	Rebuild and compare hashes	Fail page if any role projection differs
Credential status	Show last rotation and expiry, not secret	Never reveal password
Export QA report	Create manifest, counts, hashes and failed checks	Fabricated label on every page

SECTION 2

240 fabricated longitudinal patients

The data pack is an authored test population. A fixed manifest defines people and assignments. A deterministic generator creates six months of events that can be replayed exactly.

DESIGN RULE	**Action first. Meaning second. Evidence third. Raw records last.**

2.1 Regional allocation and source of truth

Census region	Profiles	Example states	Demo organizations	Clinicians
Northeast	60	MA, NY, PA, NJ, CT, ME	NE Care Network A and B	NE-C1 to NE-C3
Midwest	60	IL, OH, MI, MN, MO, WI	MW Care Network A and B	MW-C1 to MW-C3
South	60	TX, FL, GA, NC, VA, TN	SO Care Network A and B	SO-C1 to SO-C3
West	60	AZ, CA, CO, WA, OR, NV	WE Care Network A and B	WE-C1 to WE-C3
Total	240	24 displayed states	8 organizations	12 clinicians

The region field is a reporting dimension, not an estimate of clinical need. State, urbanicity, broadband access and language are authored independently. The generator must not infer behavior from a stereotype about a region or demographic group.

The four-region structure follows U.S. Census geography. If Steady later uses partner operating regions, add a separate care_region field. Do not overwrite census_region or mix both definitions in one chart.

2.2 Coverage matrix

Age coverage

Region	18-24	25-34	35-44	45-54	55-64	65+	Total
Northeast	10	10	10	10	10	10	60
Midwest	10	10	10	10	10	10	60
South	10	10	10	10	10	10	60
West	10	10	10	10	10	10	60
Total	40	40	40	40	40	40	240

Each region contains ten people in each adult age band. This is a test-design choice, not a claim about real prevalence. It gives the demo enough cases to show age filters without making one region appear empty.

Outcome archetypes

Archetype	Count	Story generated
Early response	30	High early engagement and early observed improvement
Steady response	30	Regular use and gradual observed improvement
Late response	30	Early plateau followed by later change
No change	30	Use without material measure change
Sporadic use	30	Irregular check-ins and incomplete follow-up
Module mismatch	30	High use concentrated in a low-response module mix
Access barrier	30	Missed activity linked to authored access events
Safety pause	30	Fixed safety event, pause, review and bounded re-entry

Archetypes create testable stories. They do not label real people, predict a diagnosis or decide care.

2.3 Demographic fields and use rules

Field	Collection rule	Analytics use	Do not do
Age	Store exact fabricated birth date; display band in aggregates	Stratify access, use and outcomes	Assume age causes a result
Race	Self-described fabricated value; allow multiple values	Representation and disparity audit	Use as a direct care-selection rule
Ethnicity	Separate field from race	Representation and disparity audit	Collapse into race or infer from name
Language	Preferred language plus interpreter need	Access and engagement analysis	Treat language as ability or motivation
Sex and gender	Separate fields if included; permit unknown and decline	Access and fairness audit	Infer identity or impose binary-only values
Disability/access	Functional access needs, not labels alone	Accessibility testing and service-gap review	Reduce access based on disability
Region/state	Authored geography	Capacity, access and delivery analysis	Use geography as a proxy for race
Socioeconomic context	Authored insurance and access-barrier fields	Navigation and access analysis	Create a hidden deprivation score for person routing

Protected characteristics stay available to the controlled audit layer. Payer and organization pages receive suppressed aggregates. Clinician pages show only fields needed for respectful care and access support, with patient-reported provenance.

2.4 Longitudinal event recipe

Domain	Per-person six-month target	Generation rule
Enrollment	1 account, 1 consent set, 1 tenant assignment	Fixed on day 0 with versioned consent
Measures	4 to 8 completed, 0 to 3 partial	Scheduled cadence with archetype-specific missingness
Daily check-ins	18 to 90	Calendar generator plus access and engagement pattern
Modules	8 to 55 completed	Grounding, breathing, learning, preparation and support mix
Sessions	0 to 8 simulated support sessions	Never treated as trauma-processing proof
Safety	0 to 3 fixed gate events	Deterministic inputs produce expected gate output
Clinician actions	1 to 12	Review, message, assign, note, pause and re-entry actions
Corrections	0 to 2	Append correction and supersede prior display value
Utilization	Authored encounters and allowed cost examples	Separate observed claims from modeled cost
Audit	Every privileged read and write	Actor, purpose, subject, tenant, projection and outcome

All timestamps derive from demo_epoch plus seeded offsets. Randomness uses a documented pseudorandom generator and one stable seed per profile. Re-running the same version must produce the same event ids, timestamps, values and projection hashes.

2.5 Canonical fabricated profile schema


```
{ "person_id": "ST-WE-017", "fabricated": true, "dataset_version": "demo-population-v1", "seed": 100217, "tenant_id": "demo-org-west-01", "clinician_id": "WE-C2", "census_region": "West", "state": "AZ", "age_band": "35-44", "race": ["Multiracial"], "ethnicity": "Not Hispanic/Latino", "preferred_language": "English", "archetype": "Steady response", "baseline_measure": 18, "expected_followup_measure": 12, "expected_gate_events": 0, "expected_projection_version": "person_summary.v4" }
```


Rule	Acceptance
No real identifiers	Names, emails, addresses, phones, free text and dates are generated from demo dictionaries
No hidden linkage	No seed, hash or source field points to a production person
Provenance	fabricated=true and dataset_version appear on every root record and export
Stable ids	Ids are deterministic within a dataset version
Breaking change	Any changed generator or definition increments dataset_version
Reset	Delete only the isolated demo tenant, then rebuild from manifest

2.6 Seed manifest 1-20

ID	Reg	ST	Age	Race	Ethn	Lang	Archetype	Clin	Base	F/U	Safety
ST-NE-001	NE	MA	18-24 18	AIAN	Hisp.	English	Early response	NE-C1	15	8	No active gate
ST-NE-002	NE	NY	18-24 19	Asian	Not Hisp.	English	Steady response	NE-C2	22	18	No active gate
ST-NE-003	NE	PA	18-24 20	Black	Not Hisp.	Spanish	Late response	NE-C3	13	8	No active gate
ST-NE-004	NE	NJ	18-24 21	NHPI	Not Hisp.	English	No change	NE-C1	20	20	No active gate
ST-NE-005	NE	CT	18-24 22	White	Hisp.	Mandarin	Sporadic use	NE-C2	11	10	No active gate
ST-NE-006	NE	ME	18-24 23	Multiracial	Not Hisp.	English	Module mismatch	NE-C3	18	16	No active gate
ST-NE-007	NE	MA	18-24 24	AIAN	Not Hisp.	English	Access barrier	NE-C1	9	8	No active gate
ST-NE-008	NE	NY	18-24 18	Asian	Not Hisp.	English	Safety pause	NE-C2	16	14	Fixed pause
ST-NE-009	NE	PA	18-24 19	Black	Hisp.	Spanish	Early response	NE-C3	23	15	No active gate
ST-NE-010	NE	NJ	18-24 20	NHPI	Not Hisp.	English	Steady response	NE-C1	14	9	No active gate
ST-NE-011	NE	NY	25-34 25	Asian	Not Hisp.	Spanish	Late response	NE-C2	8	5	No active gate
ST-NE-012	NE	PA	25-34 26	Black	Not Hisp.	English	No change	NE-C3	15	14	No active gate
ST-NE-013	NE	NJ	25-34 27	NHPI	Hisp.	Mandarin	Sporadic use	NE-C1	22	20	No active gate
ST-NE-014	NE	CT	25-34 28	White	Not Hisp.	English	Module mismatch	NE-C2	13	13	No active gate
ST-NE-015	NE	ME	25-34 29	Multiracial	Not Hisp.	English	Access barrier	NE-C3	20	18	No active gate
ST-NE-016	NE	MA	25-34 30	AIAN	Not Hisp.	English	Safety pause	NE-C1	11	8	Fixed pause
ST-NE-017	NE	NY	25-34 31	Asian	Hisp.	Spanish	Early response	NE-C2	18	12	Review event
ST-NE-018	NE	PA	25-34 32	Black	Not Hisp.	English	Steady response	NE-C3	9	3	No active gate
ST-NE-019	NE	NJ	25-34 33	NHPI	Not Hisp.	Mandarin	Late response	NE-C1	16	12	No active gate
ST-NE-020	NE	CT	25-34 34	White	Not Hisp.	English	No change	NE-C2	23	24	No active gate

Base and F/U are fabricated measure values used to validate trajectory math. They are not diagnoses, treatment decisions or outcome claims.

2.6 Seed manifest 21-40

ID	Reg	ST	Age	Race	Ethn	Lang	Archetype	Clin	Base	F/U	Safety
ST-NE-021	NE	PA	35-44 35	Black	Hisp.	Mandarin	Sporadic use	NE-C3	17	14	No active gate
ST-NE-022	NE	NJ	35-44 36	NHPI	Not Hisp.	English	Module mismatch	NE-C1	8	7	No active gate
ST-NE-023	NE	CT	35-44 37	White	Not Hisp.	English	Access barrier	NE-C2	15	15	No active gate
ST-NE-024	NE	ME	35-44 38	Multiracial	Not Hisp.	English	Safety pause	NE-C3	22	18	Fixed pause
ST-NE-025	NE	MA	35-44 39	AIAN	Hisp.	Spanish	Early response	NE-C1	13	6	No active gate
ST-NE-026	NE	NY	35-44 40	Asian	Not Hisp.	English	Steady response	NE-C2	20	16	No active gate
ST-NE-027	NE	PA	35-44 41	Black	Not Hisp.	Mandarin	Late response	NE-C3	11	6	No active gate
ST-NE-028	NE	NJ	35-44 42	NHPI	Not Hisp.	English	No change	NE-C1	18	18	No active gate
ST-NE-029	NE	CT	35-44 43	White	Hisp.	English	Sporadic use	NE-C2	9	8	No active gate
ST-NE-030	NE	ME	35-44 44	Multiracial	Not Hisp.	English	Module mismatch	NE-C3	16	14	No active gate
ST-NE-031	NE	NJ	45-54 45	NHPI	Not Hisp.	English	Access barrier	NE-C1	10	9	No active gate
ST-NE-032	NE	CT	45-54 46	White	Not Hisp.	English	Safety pause	NE-C2	17	15	Fixed pause
ST-NE-033	NE	ME	45-54 47	Multiracial	Hisp.	Spanish	Early response	NE-C3	8	0	No active gate
ST-NE-034	NE	MA	45-54 48	AIAN	Not Hisp.	English	Steady response	NE-C1	15	10	Review event
ST-NE-035	NE	NY	45-54 49	Asian	Not Hisp.	Mandarin	Late response	NE-C2	22	19	No active gate
ST-NE-036	NE	PA	45-54 50	Black	Not Hisp.	English	No change	NE-C3	13	12	No active gate
ST-NE-037	NE	NJ	45-54 51	NHPI	Hisp.	English	Sporadic use	NE-C1	20	18	No active gate
ST-NE-038	NE	CT	45-54 52	White	Not Hisp.	English	Module mismatch	NE-C2	11	11	No active gate
ST-NE-039	NE	ME	45-54 53	Multiracial	Not Hisp.	Spanish	Access barrier	NE-C3	18	16	No active gate
ST-NE-040	NE	MA	45-54 54	AIAN	Not Hisp.	English	Safety pause	NE-C1	9	6	Fixed pause

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2.6 Seed manifest 41-60

ID	Reg	ST	Age	Race	Ethn	Lang	Archetype	Clin	Base	F/U	Safety
ST-NE-041	NE	CT	55-64 55	White	Hisp.	Spanish	Early response	NE-C2	19	13	No active gate
ST-NE-042	NE	ME	55-64 56	Multiracial	Not Hisp.	English	Steady response	NE-C3	10	4	No active gate
ST-NE-043	NE	MA	55-64 57	AIAN	Not Hisp.	Mandarin	Late response	NE-C1	17	13	No active gate
ST-NE-044	NE	NY	55-64 58	Asian	Not Hisp.	English	No change	NE-C2	8	9	No active gate
ST-NE-045	NE	PA	55-64 59	Black	Hisp.	English	Sporadic use	NE-C3	15	12	No active gate
ST-NE-046	NE	NJ	55-64 60	NHPI	Not Hisp.	English	Module mismatch	NE-C1	22	21	No active gate
ST-NE-047	NE	CT	55-64 61	White	Not Hisp.	Spanish	Access barrier	NE-C2	13	13	No active gate
ST-NE-048	NE	ME	55-64 62	Multiracial	Not Hisp.	English	Safety pause	NE-C3	20	16	Fixed pause
ST-NE-049	NE	MA	55-64 63	AIAN	Hisp.	Mandarin	Early response	NE-C1	11	4	No active gate
ST-NE-050	NE	NY	55-64 64	Asian	Not Hisp.	English	Steady response	NE-C2	18	14	No active gate
ST-NE-051	NE	ME	65+ 65	Multiracial	Not Hisp.	Mandarin	Late response	NE-C3	12	7	Review event
ST-NE-052	NE	MA	65+ 66	AIAN	Not Hisp.	English	No change	NE-C1	19	19	No active gate
ST-NE-053	NE	NY	65+ 67	Asian	Hisp.	English	Sporadic use	NE-C2	10	9	No active gate
ST-NE-054	NE	PA	65+ 68	Black	Not Hisp.	English	Module mismatch	NE-C3	17	15	No active gate
ST-NE-055	NE	NJ	65+ 69	NHPI	Not Hisp.	Spanish	Access barrier	NE-C1	8	7	No active gate
ST-NE-056	NE	CT	65+ 70	White	Not Hisp.	English	Safety pause	NE-C2	15	13	Fixed pause
ST-NE-057	NE	ME	65+ 71	Multiracial	Hisp.	Mandarin	Early response	NE-C3	22	14	No active gate
ST-NE-058	NE	MA	65+ 65	AIAN	Not Hisp.	English	Steady response	NE-C1	13	8	No active gate
ST-NE-059	NE	NY	65+ 66	Asian	Not Hisp.	English	Late response	NE-C2	20	17	No active gate
ST-NE-060	NE	PA	65+ 67	Black	Not Hisp.	English	No change	NE-C3	11	10	No active gate

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2.6 Seed manifest 61-80

ID	Re g	ST	Age	Race	Ethn	Lang	Archetype	Clin	Bas e	F/U	Safety
ST-MW-001	MW	IL	18-24 18	Asian	Not Hisp.	English	Sporadic use	MW-C1	19	17	No active gate
ST-MW-002	MW	OH	18-24 19	Black	Not Hisp.	Spanish	Module mismatch	MW-C2	10	10	No active gate
ST-MW-003	MW	MI	18-24 20	NHPI	Not Hisp.	English	Access barrier	MW-C3	17	15	No active gate
ST-MW-004	MW	MN	18-24 21	White	Hisp.	Mandarin	Safety pause	MW-C1	8	5	Fixed pause
ST-MW-005	MW	MO	18-24 22	Multiracial	Not Hisp.	English	Early response	MW-C2	15	9	No active gate
ST-MW-006	MW	WI	18-24 23	AIAN	Not Hisp.	English	Steady response	MW-C3	22	16	No active gate
ST-MW-007	MW	IL	18-24 24	Asian	Not Hisp.	English	Late response	MW-C1	13	9	No active gate
ST-MW-008	MW	OH	18-24 18	Black	Hisp.	Spanish	No change	MW-C2	20	21	Review event
ST-MW-009	MW	MI	18-24 19	NHPI	Not Hisp.	English	Sporadic use	MW-C3	11	8	No active gate
ST-MW-010	MW	MN	18-24 20	White	Not Hisp.	Mandarin	Module mismatch	MW-C1	18	17	No active gate
ST-MW-011	MW	OH	25-34 25	Black	Not Hisp.	English	Access barrier	MW-C2	12	12	No active gate
ST-MW-012	MW	MI	25-34 26	NHPI	Hisp.	Mandarin	Safety pause	MW-C3	19	15	Fixed pause
ST-MW-013	MW	MN	25-34 27	White	Not Hisp.	English	Early response	MW-C1	10	3	No active gate
ST-MW-014	MW	MO	25-34 28	Multiracial	Not Hisp.	English	Steady response	MW-C2	17	13	No active gate
ST-MW-015	MW	WI	25-34 29	AIAN	Not Hisp.	English	Late response	MW-C3	8	3	No active gate
ST-MW-016	MW	IL	25-34 30	Asian	Hisp.	Spanish	No change	MW-C1	15	15	No active gate
ST-MW-017	MW	OH	25-34 31	Black	Not Hisp.	English	Sporadic use	MW-C2	22	21	No active gate
ST-MW-018	MW	MI	25-34 32	NHPI	Not Hisp.	Mandarin	Module mismatch	MW-C3	13	11	No active gate
ST-MW-019	MW	MN	25-34 33	White	Not Hisp.	English	Access barrier	MW-C1	20	19	No active gate
ST-MW-020	MW	MO	25-34 34	Multiracial	Hisp.	English	Safety pause	MW-C2	11	9	Fixed pause

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2.6 Seed manifest 81-100

ID	Reg	ST	Age	Race	Ethn	Lang	Archetype	Clin	Base	F/U	Safety
ST-MW-021	MW	MI	35-44 35	NHPI	Not Hisp.	English	Early response	MW-C3	21	13	No active gate
ST-MW-022	MW	MN	35-44 36	White	Not Hisp.	English	Steady response	MW-C1	12	7	No active gate
ST-MW-023	MW	MO	35-44 37	Multiracial	Not Hisp.	English	Late response	MW-C2	19	16	No active gate
ST-MW-024	MW	WI	35-44 38	AIAN	Hisp.	Spanish	No change	MW-C3	10	9	No active gate
ST-MW-025	MW	IL	35-44 39	Asian	Not Hisp.	English	Sporadic use	MW-C1	17	15	Review event
ST-MW-026	MW	OH	35-44 40	Black	Not Hisp.	Mandarin	Module mismatch	MW-C2	8	8	No active gate
ST-MW-027	MW	MI	35-44 41	NHPI	Not Hisp.	English	Access barrier	MW-C3	15	13	No active gate
ST-MW-028	MW	MN	35-44 42	White	Hisp.	English	Safety pause	MW-C1	22	19	Fixed pause
ST-MW-029	MW	MO	35-44 43	Multiracial	Not Hisp.	English	Early response	MW-C2	13	7	No active gate
ST-MW-030	MW	WI	35-44 44	AIAN	Not Hisp.	Spanish	Steady response	MW-C3	20	14	No active gate
ST-MW-031	MW	MN	45-54 45	White	Not Hisp.	English	Late response	MW-C1	14	10	No active gate
ST-MW-032	MW	MO	45-54 46	Multiracial	Hisp.	Spanish	No change	MW-C2	21	22	No active gate
ST-MW-033	MW	WI	45-54 47	AIAN	Not Hisp.	English	Sporadic use	MW-C3	12	9	No active gate
ST-MW-034	MW	IL	45-54 48	Asian	Not Hisp.	Mandarin	Module mismatch	MW-C1	19	18	No active gate
ST-MW-035	MW	OH	45-54 49	Black	Not Hisp.	English	Access barrier	MW-C2	10	10	No active gate
ST-MW-036	MW	MI	45-54 50	NHPI	Hisp.	English	Safety pause	MW-C3	17	13	Fixed pause
ST-MW-037	MW	MN	45-54 51	White	Not Hisp.	English	Early response	MW-C1	8	1	No active gate
ST-MW-038	MW	MO	45-54 52	Multiracial	Not Hisp.	Spanish	Steady response	MW-C2	15	11	No active gate
ST-MW-039	MW	WI	45-54 53	AIAN	Not Hisp.	English	Late response	MW-C3	22	17	No active gate
ST-MW-040	MW	IL	45-54 54	Asian	Hisp.	Mandarin	No change	MW-C1	13	13	No active gate

Base and F/U are fabricated measure values used to validate trajectory math. They are not diagnoses, treatment decisions or outcome claims.

2.6 Seed manifest 101-120

ID	Reg	ST	Age	Race	Ethn	Lang	Archetype	Clin	Base	F/U	Safety
ST-MW-041	MW	MO	55-64 55	Multiracial	Not Hisp.	English	Sporadic use	MW-C2	23	22	No active gate
ST-MW-042	MW	WI	55-64 56	AIAN	Not Hisp.	Mandarin	Module mismatch	MW-C3	14	12	Review event
ST-MW-043	MW	IL	55-64 57	Asian	Not Hisp.	English	Access barrier	MW-C1	21	20	No active gate
ST-MW-044	MW	OH	55-64 58	Black	Hisp.	English	Safety pause	MW-C2	12	10	Fixed pause
ST-MW-045	MW	MI	55-64 59	NHPI	Not Hisp.	English	Early response	MW-C3	19	11	No active gate
ST-MW-046	MW	MN	55-64 60	White	Not Hisp.	Spanish	Steady response	MW-C1	10	5	No active gate
ST-MW-047	MW	MO	55-64 61	Multiracial	Not Hisp.	English	Late response	MW-C2	17	14	No active gate
ST-MW-048	MW	WI	55-64 62	AIAN	Hisp.	Mandarin	No change	MW-C3	8	7	No active gate
ST-MW-049	MW	IL	55-64 63	Asian	Not Hisp.	English	Sporadic use	MW-C1	15	13	No active gate
ST-MW-050	MW	OH	55-64 64	Black	Not Hisp.	English	Module mismatch	MW-C2	22	22	No active gate
ST-MW-051	MW	WI	65+ 65	AIAN	Not Hisp.	English	Access barrier	MW-C3	16	14	No active gate
ST-MW-052	MW	IL	65+ 66	Asian	Hisp.	English	Safety pause	MW-C1	23	20	Fixed pause
ST-MW-053	MW	OH	65+ 67	Black	Not Hisp.	English	Early response	MW-C2	14	8	No active gate
ST-MW-054	MW	MI	65+ 68	NHPI	Not Hisp.	Spanish	Steady response	MW-C3	21	15	No active gate
ST-MW-055	MW	MN	65+ 69	White	Not Hisp.	English	Late response	MW-C1	12	8	No active gate
ST-MW-056	MW	MO	65+ 70	Multiracial	Hisp.	Mandarin	No change	MW-C2	19	20	No active gate
ST-MW-057	MW	WI	65+ 71	AIAN	Not Hisp.	English	Sporadic use	MW-C3	10	7	No active gate
ST-MW-058	MW	IL	65+ 65	Asian	Not Hisp.	English	Module mismatch	MW-C1	17	16	No active gate
ST-MW-059	MW	OH	65+ 66	Black	Not Hisp.	English	Access barrier	MW-C2	8	8	Review event
ST-MW-060	MW	MI	65+ 67	NHPI	Hisp.	Spanish	Safety pause	MW-C3	15	11	Fixed pause

Base and F/U are fabricated measure values used to validate trajectory math. They are not diagnoses, treatment decisions or outcome claims.

2.6 Seed manifest 121-140

ID	Reg	ST	Age	Race	Ethn	Lang	Archetype	Clin	Base	F/U	Safety
ST-SO-001	SO	TX	18-24 18	Black	Not Hisp.	Spanish	Early response	SO-C1	23	16	No active gate
ST-SO-002	SO	FL	18-24 19	NHPI	Not Hisp.	English	Steady response	SO-C2	14	10	No active gate
ST-SO-003	SO	GA	18-24 20	White	Hisp.	Mandarin	Late response	SO-C3	21	16	No active gate
ST-SO-004	SO	NC	18-24 21	Multiracial	Not Hisp.	English	No change	SO-C1	12	12	No active gate
ST-SO-005	SO	VA	18-24 22	AIAN	Not Hisp.	English	Sporadic use	SO-C2	19	18	No active gate
ST-SO-006	SO	TN	18-24 23	Asian	Not Hisp.	English	Module mismatch	SO-C3	10	8	No active gate
ST-SO-007	SO	TX	18-24 24	Black	Hisp.	Spanish	Access barrier	SO-C1	17	16	No active gate
ST-SO-008	SO	FL	18-24 18	NHPI	Not Hisp.	English	Safety pause	SO-C2	8	6	Fixed pause
ST-SO-009	SO	GA	18-24 19	White	Not Hisp.	Mandarin	Early response	SO-C3	15	7	No active gate
ST-SO-010	SO	NC	18-24 20	Multiracial	Not Hisp.	English	Steady response	SO-C1	22	17	No active gate
ST-SO-011	SO	FL	25-34 25	NHPI	Hisp.	Mandarin	Late response	SO-C2	16	13	No active gate
ST-SO-012	SO	GA	25-34 26	White	Not Hisp.	English	No change	SO-C3	23	22	No active gate
ST-SO-013	SO	NC	25-34 27	Multiracial	Not Hisp.	English	Sporadic use	SO-C1	14	12	No active gate
ST-SO-014	SO	VA	25-34 28	AIAN	Not Hisp.	English	Module mismatch	SO-C2	21	21	No active gate
ST-SO-015	SO	TN	25-34 29	Asian	Hisp.	Spanish	Access barrier	SO-C3	12	10	No active gate
ST-SO-016	SO	TX	25-34 30	Black	Not Hisp.	English	Safety pause	SO-C1	19	16	Fixed pause
ST-SO-017	SO	FL	25-34 31	NHPI	Not Hisp.	Mandarin	Early response	SO-C2	10	4	No active gate
ST-SO-018	SO	GA	25-34 32	White	Not Hisp.	English	Steady response	SO-C3	17	11	No active gate
ST-SO-019	SO	NC	25-34 33	Multiracial	Hisp.	English	Late response	SO-C1	8	4	No active gate
ST-SO-020	SO	VA	25-34 34	AIAN	Not Hisp.	English	No change	SO-C2	15	16	No active gate

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2.6 Seed manifest 141-160

ID	Re g	ST	Age	Race	Ethn	Lang	Archetype	Clin	Bas e	F/U	Safety
ST-SO-021	SO	GA	35-44 35	White	Not Hisp.	English	Sporadic use	SO-C3	9	6	No active gate
ST-SO-022	SO	NC	35-44 36	Multiracial	Not Hisp.	English	Module mismatch	SO-C1	16	15	No active gate
ST-SO-023	SO	VA	35-44 37	AIAN	Hisp.	Spanish	Access barrier	SO-C2	23	23	No active gate
ST-SO-024	SO	TN	35-44 38	Asian	Not Hisp.	English	Safety pause	SO-C3	14	10	Fixed pause
ST-SO-025	SO	TX	35-44 39	Black	Not Hisp.	Mandarin	Early response	SO-C1	21	14	No active gate
ST-SO-026	SO	FL	35-44 40	NHPI	Not Hisp.	English	Steady response	SO-C2	12	8	No active gate
ST-SO-027	SO	GA	35-44 41	White	Hisp.	English	Late response	SO-C3	19	14	No active gate
ST-SO-028	SO	NC	35-44 42	Multiracial	Not Hisp.	English	No change	SO-C1	10	10	No active gate
ST-SO-029	SO	VA	35-44 43	AIAN	Not Hisp.	Spanish	Sporadic use	SO-C2	17	16	No active gate
ST-SO-030	SO	TN	35-44 44	Asian	Not Hisp.	English	Module mismatch	SO-C3	8	6	No active gate
ST-SO-031	SO	NC	45-54 45	Multiracial	Hisp.	Spanish	Access barrier	SO-C1	18	17	No active gate
ST-SO-032	SO	VA	45-54 46	AIAN	Not Hisp.	English	Safety pause	SO-C2	9	7	Fixed pause
ST-SO-033	SO	TN	45-54 47	Asian	Not Hisp.	Mandarin	Early response	SO-C3	16	8	Review event
ST-SO-034	SO	TX	45-54 48	Black	Not Hisp.	English	Steady response	SO-C1	23	18	No active gate
ST-SO-035	SO	FL	45-54 49	NHPI	Hisp.	English	Late response	SO-C2	14	11	No active gate
ST-SO-036	SO	GA	45-54 50	White	Not Hisp.	English	No change	SO-C3	21	20	No active gate
ST-SO-037	SO	NC	45-54 51	Multiracial	Not Hisp.	Spanish	Sporadic use	SO-C1	12	10	No active gate
ST-SO-038	SO	VA	45-54 52	AIAN	Not Hisp.	English	Module mismatch	SO-C2	19	19	No active gate
ST-SO-039	SO	TN	45-54 53	Asian	Hisp.	Mandarin	Access barrier	SO-C3	10	8	No active gate
ST-SO-040	SO	TX	45-54 54	Black	Not Hisp.	English	Safety pause	SO-C1	17	14	Fixed pause

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2.6 Seed manifest 161-180

ID	Reg	ST	Age	Race	Ethn	Lang	Archetype	Clin	Base	F/U	Safety
ST-SO-041	SO	VA	55-64 55	AIAN	Not Hisp.	Mandarin	Early response	SO-C2	11	5	No active gate
ST-SO-042	SO	TN	55-64 56	Asian	Not Hisp.	English	Steady response	SO-C3	18	12	No active gate
ST-SO-043	SO	TX	55-64 57	Black	Hisp.	English	Late response	SO-C1	9	5	No active gate
ST-SO-044	SO	FL	55-64 58	NHPI	Not Hisp.	English	No change	SO-C2	16	17	No active gate
ST-SO-045	SO	GA	55-64 59	White	Not Hisp.	Spanish	Sporadic use	SO-C3	23	20	No active gate
ST-SO-046	SO	NC	55-64 60	Multiracial	Not Hisp.	English	Module mismatch	SO-C1	14	13	No active gate
ST-SO-047	SO	VA	55-64 61	AIAN	Hisp.	Mandarin	Access barrier	SO-C2	21	21	No active gate
ST-SO-048	SO	TN	55-64 62	Asian	Not Hisp.	English	Safety pause	SO-C3	12	8	Fixed pause
ST-SO-049	SO	TX	55-64 63	Black	Not Hisp.	English	Early response	SO-C1	19	12	No active gate
ST-SO-050	SO	FL	55-64 64	NHPI	Not Hisp.	English	Steady response	SO-C2	10	6	Review event
ST-SO-051	SO	TN	65+ 65	Asian	Hisp.	English	Late response	SO-C3	20	15	No active gate
ST-SO-052	SO	TX	65+ 66	Black	Not Hisp.	English	No change	SO-C1	11	11	No active gate
ST-SO-053	SO	FL	65+ 67	NHPI	Not Hisp.	Spanish	Sporadic use	SO-C2	18	17	No active gate
ST-SO-054	SO	GA	65+ 68	White	Not Hisp.	English	Module mismatch	SO-C3	9	7	No active gate
ST-SO-055	SO	NC	65+ 69	Multiracial	Hisp.	Mandarin	Access barrier	SO-C1	16	15	No active gate
ST-SO-056	SO	VA	65+ 70	AIAN	Not Hisp.	English	Safety pause	SO-C2	23	21	Fixed pause
ST-SO-057	SO	TN	65+ 71	Asian	Not Hisp.	English	Early response	SO-C3	14	6	No active gate
ST-SO-058	SO	TX	65+ 65	Black	Not Hisp.	English	Steady response	SO-C1	21	16	No active gate
ST-SO-059	SO	FL	65+ 66	NHPI	Hisp.	Spanish	Late response	SO-C2	12	9	No active gate
ST-SO-060	SO	GA	65+ 67	White	Not Hisp.	English	No change	SO-C3	19	18	No active gate

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2.6 Seed manifest 181-200

ID	Reg	ST	Age	Race	Ethn	Lang	Archetype	Clin	Base	F/U	Safety
ST-WE-001	WE	AZ	18-24 18	NHPI	Not Hisp.	English	Sporadic use	WE-C1	11	9	No active gate
ST-WE-002	WE	CA	18-24 19	White	Hisp.	Mandarin	Module mismatch	WE-C2	18	18	No active gate
ST-WE-003	WE	CO	18-24 20	Multiracial	Not Hisp.	English	Access barrier	WE-C3	9	7	No active gate
ST-WE-004	WE	WA	18-24 21	AIAN	Not Hisp.	English	Safety pause	WE-C1	16	13	Fixed pause
ST-WE-005	WE	OR	18-24 22	Asian	Not Hisp.	English	Early response	WE-C2	23	17	No active gate
ST-WE-006	WE	NV	18-24 23	Black	Hisp.	Spanish	Steady response	WE-C3	14	8	No active gate
ST-WE-007	WE	AZ	18-24 24	NHPI	Not Hisp.	English	Late response	WE-C1	21	17	Review event
ST-WE-008	WE	CA	18-24 18	White	Not Hisp.	Mandarin	No change	WE-C2	12	13	No active gate
ST-WE-009	WE	CO	18-24 19	Multiracial	Not Hisp.	English	Sporadic use	WE-C3	19	16	No active gate
ST-WE-010	WE	WA	18-24 20	AIAN	Hisp.	English	Module mismatch	WE-C1	10	9	No active gate
ST-WE-011	WE	CA	25-34 25	White	Not Hisp.	English	Access barrier	WE-C2	20	20	No active gate
ST-WE-012	WE	CO	25-34 26	Multiracial	Not Hisp.	English	Safety pause	WE-C3	11	7	Fixed pause
ST-WE-013	WE	WA	25-34 27	AIAN	Not Hisp.	English	Early response	WE-C1	18	11	No active gate
ST-WE-014	WE	OR	25-34 28	Asian	Hisp.	Spanish	Steady response	WE-C2	9	5	No active gate
ST-WE-015	WE	NV	25-34 29	Black	Not Hisp.	English	Late response	WE-C3	16	11	No active gate
ST-WE-016	WE	AZ	25-34 30	NHPI	Not Hisp.	Mandarin	No change	WE-C1	23	23	No active gate
ST-WE-017	WE	CA	25-34 31	White	Not Hisp.	English	Sporadic use	WE-C2	14	13	No active gate
ST-WE-018	WE	CO	25-34 32	Multiracial	Hisp.	English	Module mismatch	WE-C3	21	19	No active gate
ST-WE-019	WE	WA	25-34 33	AIAN	Not Hisp.	English	Access barrier	WE-C1	12	11	No active gate
ST-WE-020	WE	OR	25-34 34	Asian	Not Hisp.	Spanish	Safety pause	WE-C2	19	17	Fixed pause

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2.6 Seed manifest 201-220

ID	Reg	ST	Age	Race	Ethn	Lang	Archetype	Clin	Base	F/U	Safety
ST-WE-021	WE	CO	35-44 35	Multiracial	Not Hisp.	English	Early response	WE-C3	13	5	No active gate
ST-WE-022	WE	WA	35-44 36	AIAN	Hisp.	Spanish	Steady response	WE-C1	20	15	No active gate
ST-WE-023	WE	OR	35-44 37	Asian	Not Hisp.	English	Late response	WE-C2	11	8	No active gate
ST-WE-024	WE	NV	35-44 38	Black	Not Hisp.	Mandarin	No change	WE-C3	18	17	Review event
ST-WE-025	WE	AZ	35-44 39	NHPI	Not Hisp.	English	Sporadic use	WE-C1	9	7	No active gate
ST-WE-026	WE	CA	35-44 40	White	Hisp.	English	Module mismatch	WE-C2	16	16	No active gate
ST-WE-027	WE	CO	35-44 41	Multiracial	Not Hisp.	English	Access barrier	WE-C3	23	21	No active gate
ST-WE-028	WE	WA	35-44 42	AIAN	Not Hisp.	Spanish	Safety pause	WE-C1	14	11	Fixed pause
ST-WE-029	WE	OR	35-44 43	Asian	Not Hisp.	English	Early response	WE-C2	21	15	No active gate
ST-WE-030	WE	NV	35-44 44	Black	Hisp.	Mandarin	Steady response	WE-C3	12	6	No active gate
ST-WE-031	WE	WA	45-54 45	AIAN	Not Hisp.	English	Late response	WE-C1	22	18	No active gate
ST-WE-032	WE	OR	45-54 46	Asian	Not Hisp.	Mandarin	No change	WE-C2	13	14	No active gate
ST-WE-033	WE	NV	45-54 47	Black	Not Hisp.	English	Sporadic use	WE-C3	20	17	No active gate
ST-WE-034	WE	AZ	45-54 48	NHPI	Hisp.	English	Module mismatch	WE-C1	11	10	No active gate
ST-WE-035	WE	CA	45-54 49	White	Not Hisp.	English	Access barrier	WE-C2	18	18	No active gate
ST-WE-036	WE	CO	45-54 50	Multiracial	Not Hisp.	Spanish	Safety pause	WE-C3	9	5	Fixed pause
ST-WE-037	WE	WA	45-54 51	AIAN	Not Hisp.	English	Early response	WE-C1	16	9	No active gate
ST-WE-038	WE	OR	45-54 52	Asian	Hisp.	Mandarin	Steady response	WE-C2	23	19	No active gate
ST-WE-039	WE	NV	45-54 53	Black	Not Hisp.	English	Late response	WE-C3	14	9	No active gate
ST-WE-040	WE	AZ	45-54 54	NHPI	Not Hisp.	English	No change	WE-C1	21	21	No active gate

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2.6 Seed manifest 221-240

ID	Reg	ST	Age	Race	Ethn	Lang	Archetype	Clin	Base	F/U	Safety
ST-WE-041	WE	OR	55-64 55	Asian	Not Hisp.	English	Sporadic use	WE-C2	15	14	Review event
ST-WE-042	WE	NV	55-64 56	Black	Hisp.	English	Module mismatch	WE-C3	22	20	No active gate
ST-WE-043	WE	AZ	55-64 57	NHPI	Not Hisp.	English	Access barrier	WE-C1	13	12	No active gate
ST-WE-044	WE	CA	55-64 58	White	Not Hisp.	Spanish	Safety pause	WE-C2	20	18	Fixed pause
ST-WE-045	WE	CO	55-64 59	Multiracial	Not Hisp.	English	Early response	WE-C3	11	3	No active gate
ST-WE-046	WE	WA	55-64 60	AIAN	Hisp.	Mandarin	Steady response	WE-C1	18	13	No active gate
ST-WE-047	WE	OR	55-64 61	Asian	Not Hisp.	English	Late response	WE-C2	9	6	No active gate
ST-WE-048	WE	NV	55-64 62	Black	Not Hisp.	English	No change	WE-C3	16	15	No active gate
ST-WE-049	WE	AZ	55-64 63	NHPI	Not Hisp.	English	Sporadic use	WE-C1	23	21	No active gate
ST-WE-050	WE	CA	55-64 64	White	Hisp.	Spanish	Module mismatch	WE-C2	14	14	No active gate
ST-WE-051	WE	NV	65+ 65	Black	Not Hisp.	English	Access barrier	WE-C3	8	6	No active gate
ST-WE-052	WE	AZ	65+ 66	NHPI	Not Hisp.	Spanish	Safety pause	WE-C1	15	12	Fixed pause
ST-WE-053	WE	CA	65+ 67	White	Not Hisp.	English	Early response	WE-C2	22	16	No active gate
ST-WE-054	WE	CO	65+ 68	Multiracial	Hisp.	Mandarin	Steady response	WE-C3	13	7	No active gate
ST-WE-055	WE	WA	65+ 69	AIAN	Not Hisp.	English	Late response	WE-C1	20	16	No active gate
ST-WE-056	WE	OR	65+ 70	Asian	Not Hisp.	English	No change	WE-C2	11	12	No active gate
ST-WE-057	WE	NV	65+ 71	Black	Not Hisp.	English	Sporadic use	WE-C3	18	15	No active gate
ST-WE-058	WE	AZ	65+ 65	NHPI	Hisp.	Spanish	Module mismatch	WE-C1	9	8	Review event
ST-WE-059	WE	CA	65+ 66	White	Not Hisp.	English	Access barrier	WE-C2	16	16	No active gate
ST-WE-060	WE	CO	65+ 67	Multiracial	Not Hisp.	Mandarin	Safety pause	WE-C3	23	19	Fixed pause

Base and F/U are fabricated measure values used to validate trajectory math. They are not diagnoses, treatment decisions or outcome claims.

2.7 Event generator logic

```
for profile in manifest: rng = StableRandom(profile.seed, dataset_version) append(enrollment_event(profile))
append(consent_event(version="demo-consent-v1")) for day in demo_calendar(180): state = archetype_state(profile.archetype, day, rng)
if check_in_due(state): append(check_in_event(state)) if module_selected(state): append(module_event(state)) if measure_due(day):
append(measure_event(state, allow_partial=True)) gate = deterministic_gate(current_inputs(profile, state)) if gate.changed:
append(gate_event(gate)) if clinician_action_due(state, gate): append(clinician_action(state, gate))
append(quality_manifest(profile, expected_counts_and_hashes())) rebuild_all_projections() assert projection_hashes ==
expected_hashes
```

Generator constraints

- Do not sample outcomes independently from event history. Follow-up values must match the authored archetype and activity path.
- Do not make protected status determine success. Use constrained assignment so archetypes rotate across groups and regions.
- Create missingness intentionally and record why the value is absent: not due, skipped, declined, interrupted, failed or unavailable.
- Use separate dictionaries for supportive free text, operational notes and clinician comments. Never ask a language model to invent uncontrolled clinical narratives at runtime.
- Include edge cases: duplicate event, late arrival, correction, stale projection, partial measure, revoked consent and cross-tenant request.

2.8 Data quality manifest and reset contract

Check	Expected
Profile count	240 exactly
Regional count	60 in each of four regions
Age count	40 in each of six bands; 10 per band per region
Archetype count	30 in each of eight patterns
Fabricated flag	100 percent on root data, events, projections and exports
Orphan events	0
Cross-tenant references	0
Projection rebuild	All expected hashes match
Safety fixtures	Every fixed gate scenario returns its expected state
Small-cell output	Counts 1 through 10 suppressed on aggregate external views
Reset duration	Target under 120 seconds in demo infrastructure
Nightly reset	Success, count summary, seed version and hash recorded

The admin page blocks external demonstrations when the latest reset or projection verification failed. A presenter must never repair the demo by editing database rows directly.

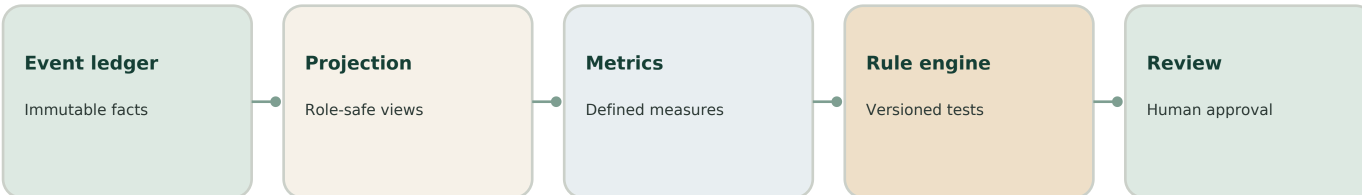
SECTION 3

Population analytics and deterministic planning

Steady converts longitudinal events into defined measures, then applies versioned rules that surface planning hypotheses for human review. The system learns at the program level before it ever predicts at the person level.

DESIGN RULE	Action first. Meaning second. Evidence third. Raw records last.

3.1 Planning engine mandate



Protected attributes audit access and outcomes. They do not select person-level care.

The first planning engine is not a treatment recommender. It produces inspectable statements such as: 'Adults 55-64 in the South completed follow-up measures less often than the overall cohort; investigate access and reminder timing.' Or: 'Grounding module completion is associated with better seven-day distress change in three regions; propose a controlled pilot.'

Every statement must show population, time window, inclusion and exclusion rules, numerator, denominator, missingness, observed versus modeled status, rule version, data version, refresh time and evidence links.

Allowed outputs

- Operational gap
- Access disparity
- Engagement pattern
- Observed outcome association
- Module-utilization pattern
- Capacity issue
- Candidate pilot hypothesis

Blocked outputs

- Diagnosis
- Person-level treatment selection based on race, age or region
- Automatic gate clearance
- Causal claim from observational data
- Payer denial or coverage restriction
- Autonomous clinical plan change

3.2 Metric dictionary

Metric	Definition	Required display
Activation	Enrolled people with first completed action within 7 days / eligible enrolled	n/N, 7-day window, exclusions
Weekly engagement	People with at least one meaningful action in week / active eligible people	n/N, action definition, missingness
Module completion	Completed module instances / started module instances	Starts, completes, abandonment, module version
Follow-up completion	Completed follow-up measure / due follow-up measures	Due, complete, partial, declined, unavailable
Observed change	Follow-up value minus baseline among paired observations	Paired n, mean, median, interval, instrument version
Responder rate	Paired people meeting configured change threshold / paired people	Threshold, n/N, interval, not diagnosis
Safety pause rate	People with fixed pause / active people	n/N, rule version; never a predicted risk score
Time to review	Elapsed time from fixed review event to documented response	Median, percentile, coverage schedule
Retention	People active at day 30, 60, 90 and 180 / eligible starters	Cohort start, censoring, n/N
Estimated cost	Modeled amount from named assumptions	Observed and modeled values shown separately

3.3 Analytic grain and cohorts

Question	Grain	Cohort key	Primary comparison
Who is using Steady?	Person-week	region, tenant, age band, race, ethnicity, language	Eligible population
Which modules are used?	Module instance	module version, person attributes, week	Starts and completions
What changed?	Paired measure episode	instrument, baseline window, follow-up window	Within-person change
Where are access gaps?	Referral or enrollment episode	region, organization, language, access need	Stage conversion
How quickly does review happen?	Safety review episode	rule, region, coverage pool	Time to documented action
Which patterns warrant a pilot?	Cohort-window	module exposure, outcome, confounder set	Adjusted and unadjusted estimates

Cohort versioning

- Save every cohort definition as executable JSON with an immutable version.
- Resolve eligibility before group filters. Never remove non-users from the denominator of an engagement rate.
- Keep partial, missing and censored records visible as separate states.
- Do not reuse one denominator across measures unless the definitions are identical.
- A dashboard link opens the exact cohort definition and sample event lineage.

3.4 Deterministic signal rules

Rule	Trigger	Output	No output when
ACCESS_GAP	Stage conversion differs by at least 10 percentage points from reference for two windows	Investigate stage, channel and owner	Cell suppressed, missingness high or denominator below threshold
FOLLOWUP_GAP	Follow-up completion is at least 12 points below reference	Review reminder timing and access barriers	Due-date logic differs across groups
MODULE_SIGNAL	Paired observed change differs by configured amount and repeats in two windows	Propose controlled pilot	Exposure definition changed or confidence interval crosses zero
REGION_CAPACITY	Demand exceeds open first-visit capacity by configured ratio	Operational capacity review	Demand or slot data stale
FAIRNESS_ALERT	Outcome, access or error-rate disparity exceeds policy threshold	Human fairness review	Protected-group completeness below policy threshold
SAFETY_REVIEW_LOAD	Fixed review events exceed staffed review capacity	Coverage and workflow review	Event classification or coverage schedule missing
DATA_QUALITY	Missingness, drift or projection mismatch crosses limit	Block planning release	Never bypassed

Thresholds shown here are product defaults for testing, not validated clinical cutoffs. Store every threshold in policy configuration, attach its owner and approval date, and prevent quiet edits.

3.5 Planning-signal state machine

State	Entry	Allowed action	Exit
Draft	Rule fires on verified metrics	Inspect cohort, lineage, missingness and confounders	Reject, request analysis or advance
Analysis requested	Owner asks for adjusted or sensitivity analysis	Run predeclared analysis plan	Return to draft with evidence
Clinical review	Signal may affect program content	Clinical reviewer comments and sets limits	Reject or advance
Fairness review	Protected-group impact or disparity is present	Assess benefit, harm, access and alternatives	Reject, mitigate or advance
Pilot proposed	Question and test design approved	Create bounded pilot protocol	Approve, revise or reject
Pilot active	Separate release flag and enrolled cohort	Monitor safety, quality and stop rules	Stop or complete
Decision recorded	Pilot result reviewed	Document adopt, do not adopt or retest	Archive with versions
Retired	Definition or evidence obsolete	Read only	No reactivation; create new version

No state transition changes a patient's permitted activity. Safety and care actions stay in their existing deterministic and clinician-governed systems.

3.6 Association, adjustment and causal restraint

A module may look successful because more engaged people choose it, because clinicians assign it to different patients, because missing follow-up differs, or because the module changed during the period. Raw averages cannot answer what caused the result.

Release ladder

Level	Method	Permitted wording	Decision use
1 Descriptive	Counts, rates, paired change, missingness	Observed among this cohort	Dashboard and gap finding
2 Stratified	Predeclared age, region and demographic strata	Observed within these strata	Fairness and consistency review
3 Adjusted	Regression or weighting with documented confounders	Adjusted association	Pilot prioritization only
4 Quasi-experimental	Difference-in-differences, interrupted time series or matched design	Estimated effect under stated assumptions	Governed program decision
5 Controlled pilot	Randomized or controlled rollout with protocol	Pilot effect estimate	Adopt, reject or retest after review

Do not use race correction factors. Protected attributes may be used to audit performance, study disparate impact and verify representation. Any future adjusted model needs a source-attribute record, fairness evaluation, monitoring plan, named owner and retirement criteria.

3.7 Fairness and small-cell controls

Control	Implementation
External small-cell suppression	Suppress counts from 1 through 10 and complementary cells that would reveal them
Internal minimum analysis size	Default n at least 30 per reported comparison; policy may require more
Missing demographic status	Show unknown, declined and missing separately; do not redistribute
Intersection checks	Audit age by race, region by language and other predeclared intersections when sample allows
Multiple testing	Control false discovery or label the view exploratory
Error analysis	Compare missingness, classification, false alarms and failed follow-up across groups
Proxy review	Test language, ZIP-derived variables, payer and utilization features for proxy behavior
Benefit and harm	Review both who gains and who may lose access or receive additional burden
Human challenge	Reviewer can reject the signal, record reasoning and request a new definition
Versioning	Data, cohort, metric, rule, model and UI explanation versions travel together

Federal health nondiscrimination rules prohibit discriminatory use of patient care decision-support tools, and the rule text describes an ongoing duty to identify tools that use protected factors. Build the audit path now, even while the first engine remains descriptive and deterministic.

3.8 Model registry for later shadow-mode work

Registry field	Required content
model_id and version	Immutable identifier and semantic version
intended use	Question, user, workflow and allowed output
out-of-scope use	Explicit prohibited decisions and populations
training data	Origin, dates, cohort, representation, exclusions and rights
features	Definitions, provenance, protected attributes and proxy review
target	Outcome definition, horizon and censoring
validation	Temporal, geographic, tenant and subgroup results
fairness	Chosen metrics, thresholds, intervals and limitations
monitoring	Drift, missingness, calibration, error rates and alert owners
human factors	How users interpret, challenge and override output
retirement	Stop conditions, replacement plan and archive rules

A registered model may generate shadow outputs for evaluation. The application does not show those outputs to patients or use them for care until Steady completes clinical, fairness, security and governance review.

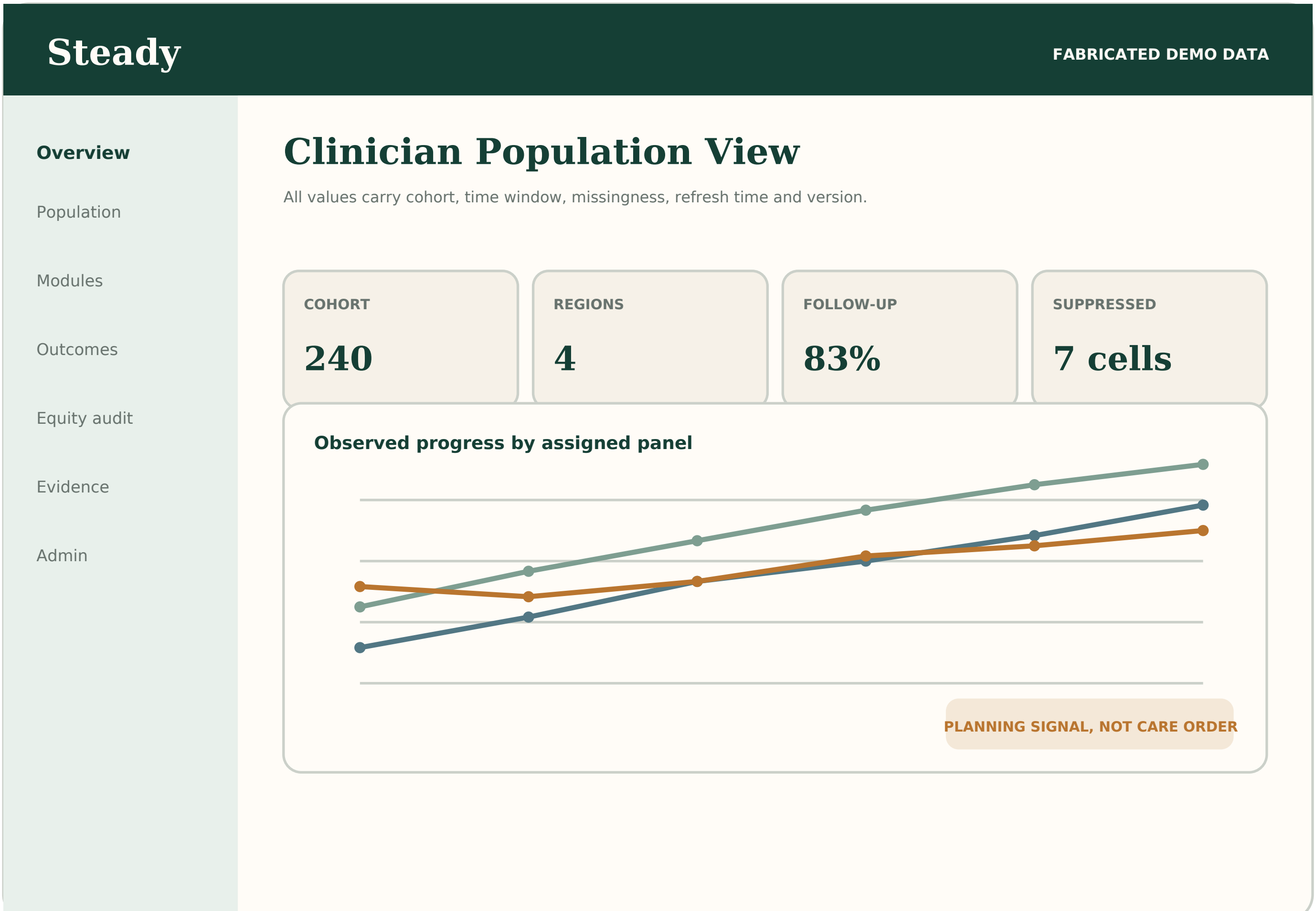
SECTION 4

Screens and interaction contracts

The interface makes role, cohort, evidence and limits obvious. Every aggregate opens to its definition. Every planning signal opens to lineage and review state.

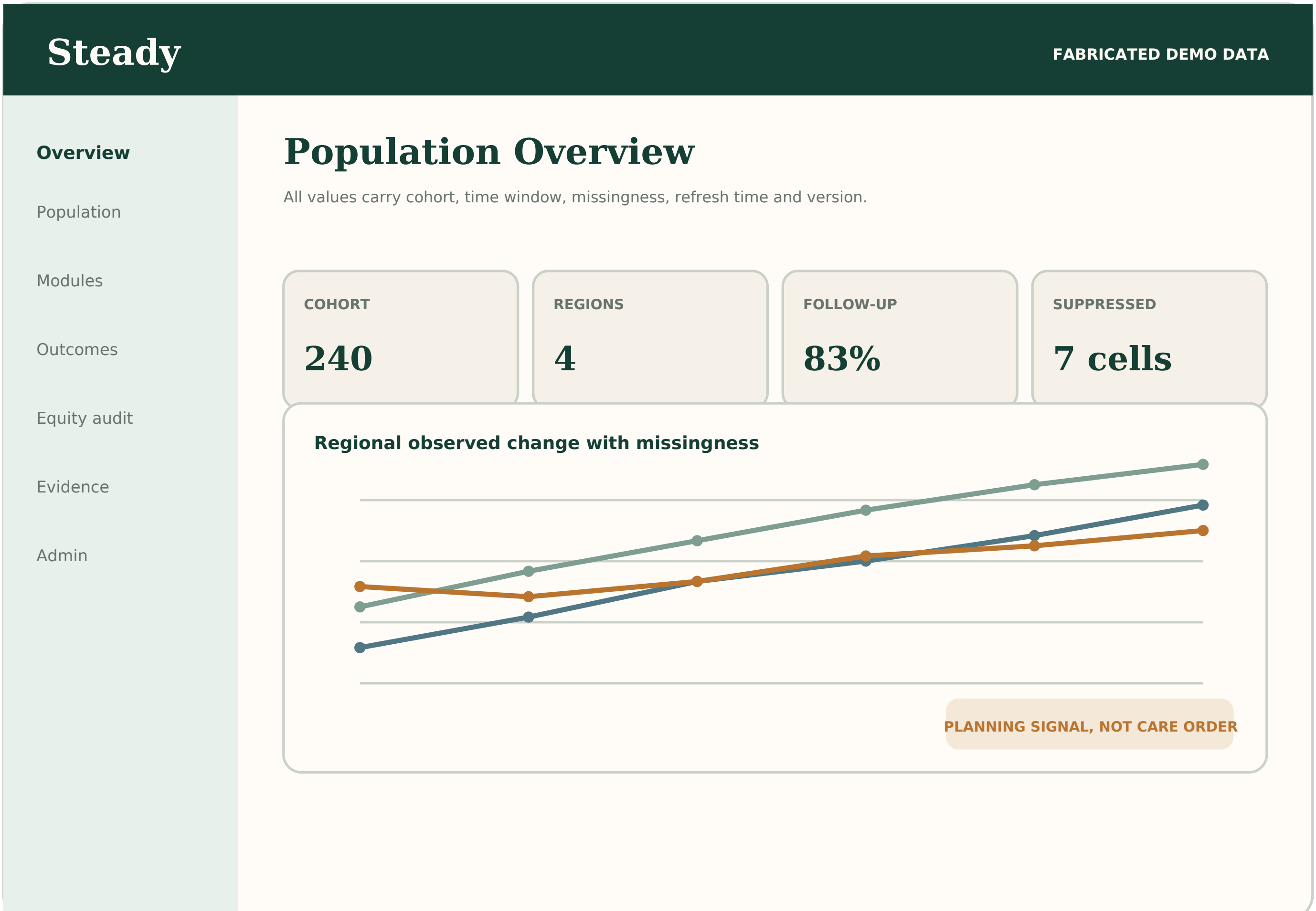
DESIGN RULE	**Action first. Meaning second. Evidence third. Raw records last.**

4.1 Clinician demo dashboard



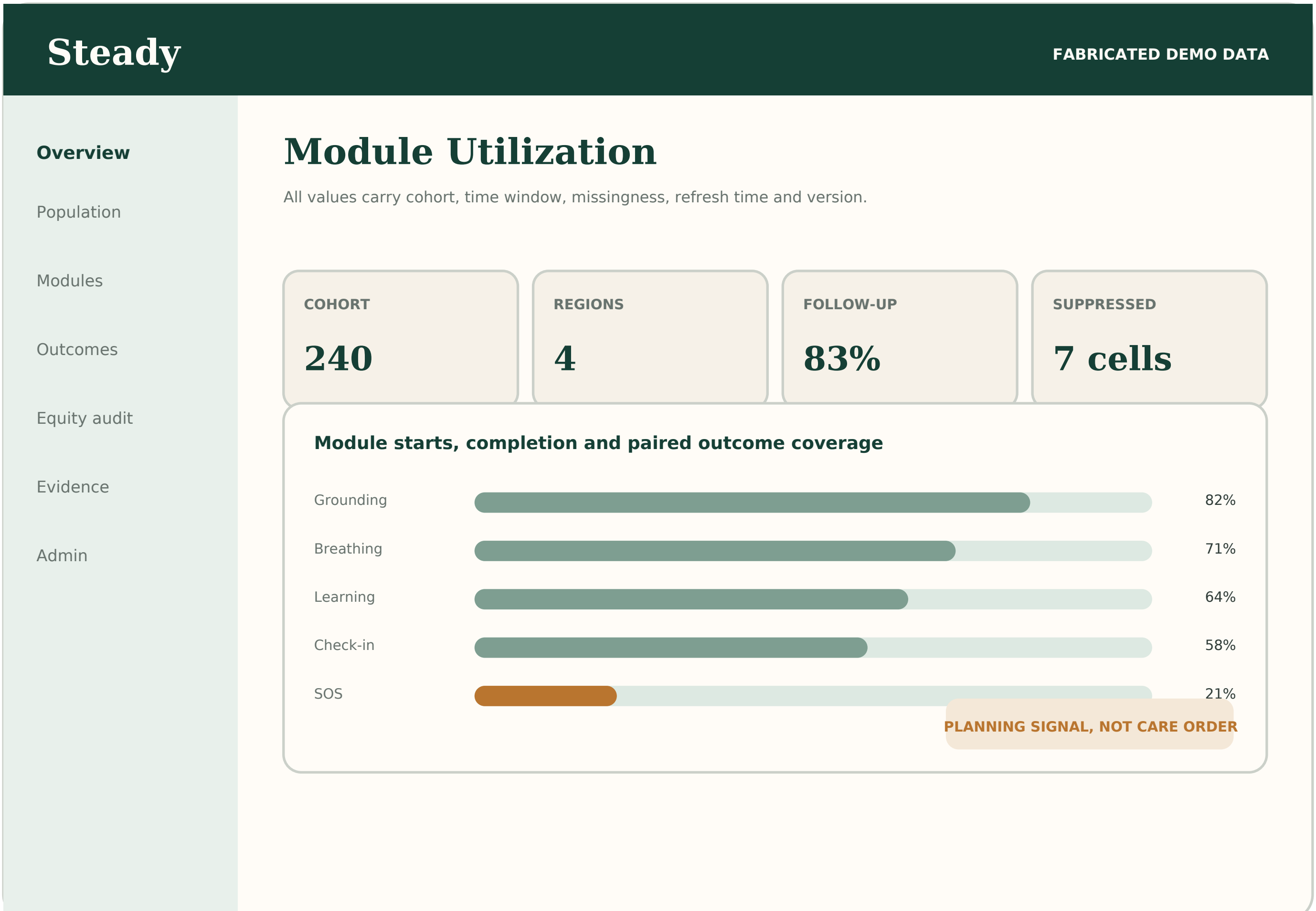
Above fold	Why
Safety review queue	Immediate fixed-state work
Assigned panel change	Who changed and freshness
Measure completion	Shows whether outcome views have coverage
Module use	Connects activity to patient history
Planning signals	Read-only aggregate hypotheses; never mixed into patient orders

4.2 Organization and payer population view



Organization asks	Payer asks	Shared contract
Are people reached?	Who is reached?	Cohort, n/N, conversion stages
Where are service gaps?	How does access vary?	Region, language and access views
Do people remain engaged?	What changes over time?	Retention, module use and paired results
Where is capacity tight?	What is utilization and cost?	Observed utilization separate from modeled cost
Which pilot should we test?	Is evidence sufficient?	Signal state, evidence, limitations and owner

4.3 Module utilization explorer



Filter	Options
Population	Tenant, organization, payer, region, assigned panel
Time	Enrollment cohort or calendar window
Module	Type, content version, session length and delivery mode
Demographic audit	Age band, race, ethnicity, language, access need
Outcome	Paired instrument, distress change, engagement, retention
Quality	Missingness, stale data, partial records, suppressed cells
Status	Observed, adjusted association, pilot or modeled

4.4 Fairness audit screen

Panel	Required content
Question	Exact metric and reason for comparing groups
Representation	Eligible, included, excluded, missing and declined counts
Performance	Point estimate, interval, n/N and reference
Missingness	Outcome and feature missingness by group
Errors	False alarm, failed projection, stale data and workflow error rates
Intersection	Predeclared two-factor views where sample permits
Small cells	Suppressed values and explanation
Decision	No issue, monitor, investigate, mitigate, block or retire
Review trail	Owner, clinical reviewer, fairness reviewer, comments and date

The screen must make it easier to discover uneven access or harm, not easier to stereotype a group. Do not rank races, assign grades to demographic groups or use red and green labels on protected identities.

4.5 Planning-signal detail screen

Section	Contents
Statement	Plain-language signal with observed or modeled label
Why it fired	Rule id, threshold, current value and prior windows
Population	Executable cohort definition and exclusions
Evidence	Charts, tables and event-lineage samples
Alternative explanations	Missingness, selection, confounding, version changes and capacity
Fairness	Protected-group coverage and disparity review
Allowed next actions	Reject, request analysis, assign owner, propose pilot
Blocked actions	No direct patient assignment, no gate change, no payer restriction
Audit	Every view, comment, state change and export

Required phrase on every candidate signal

This is a planning hypothesis based on the stated cohort and data window. It is not a diagnosis, treatment order or proof that the observed factor caused the result.

SECTION 5

Engineering contracts

The demo reuses the permanent event, tenant, consent and projection architecture. New code adds demo identities, a seed manifest, generators, aggregate metrics and planning-review objects.

DESIGN RULE	**Action first. Meaning second. Evidence third. Raw records last.**

5.1 Service boundaries

Service	Owns	Does not own
Demo identity	Demo accounts, persona catalog, session issue and revocation	Client-side authorization
Seed registry	Manifest, dataset versions, stable seeds and expected hashes	Runtime clinical decisions
Synthetic generator	Fabricated events and QA manifest	Production migration or de-identification
Event ledger	Immutable facts, corrections and replay	Role-specific display logic
Projection service	Patient, clinician, reviewer, organization, payer and admin views	Metric definitions
Metric service	Cohorts, denominators, measures, missingness and intervals	Care actions
Planning rule engine	Versioned aggregate triggers and signal lifecycle	Safety gates or person routing
Model registry	Shadow model metadata, evaluations and monitoring	Unregistered model execution
Audit service	Privileged reads, writes, exports, resets and reviews	Silent best-effort logging

5.2 Suggested API boundary

Method	Route	Purpose	Primary guard
GET	/api/demo/personas	Return enabled demo roles and labels	Demo environment only
POST	/api/demo/session	Authenticate exact role account	Identity-role match
DELETE	/api/demo/session	Revoke current session	Same session
POST	/api/demo/reset	Rebuild selected dataset version	Demo admin plus reason
GET	/api/demo/quality	Counts, hashes and failed checks	Demo admin or reviewer
GET	/api/clinician/queue	Assigned action queue	Panel relationship
GET	/api/population/metrics	Typed aggregate series	Tenant, purpose and suppression
POST	/api/population/cohorts	Save executable cohort definition	Analyst role plus schema validation
GET	/api/planning/signals	List aggregate planning signals	Role-safe aggregate access
POST	/api/planning/signals/:id/review	Record state and comments	Named reviewer and transition rule
GET	/api/planning/signals/:id/lineage	Return definitions and evidence	Minimum necessary output
GET	/api/models/registry	Read approved registry records	Reviewer, admin or allowed analyst

5.3 Aggregate metric response


```
{ "metric_id": "followup_completion.v1", "cohort_id": "south_age_55_64.v1", "window": {"start": "2026-03-01", "end": "2026-08-28"},
"grain": "person", "numerator": 31, "denominator": 40, "value": 0.775, "missing": {"partial": 2, "declined": 1, "not_due": 0},
"suppressed": false, "status": "observed", "data_version": "demo-population-v1", "metric_version": "1.0.0", "projection_version":
"population_metrics.v2", "refreshed_at": "2026-08-29T12:00:00Z", "lineage_ref": "lineage://metric-run-0092" }
```


Typed chart input

- The API returns series, units, intervals, suppression and evidence pointers. The client does not calculate clinical or business metrics from raw records.
- Observed, adjusted, pilot and modeled values use separate status fields and visual treatments.
- If a required definition, denominator, version or refresh time is missing, the chart renders a failed state rather than a number.

5.4 Planning-signal object


```
{ "signal_id": "sig-demo-0041", "signal_type": "FOLLOWUP_GAP", "state": "draft", "statement": "Follow-up completion is lower in the selected cohort.", "rule_version": "planning-rules.1.0.0", "metric_refs": ["metric-run-0092", "metric-run-0093"], "cohort_ref": "south_age_55_64.v1", "threshold": {"difference_pp": -12, "repeat_windows": 2}, "observed": {"difference_pp": -14.8, "repeat_windows": 2}, "limitations": ["fabricated data", "observational", "missing follow-up"], "allowed_actions": ["reject", "request_analysis", "assign_owner", "propose_pilot"], "blocked_actions": ["route_person", "change_gate", "deny_access"], "clinical_review": null, "fairness_review": null, "audit_ref": "audit://sig-demo-0041" }
```


The server supplies allowed_actions after policy evaluation. The client never invents or widens the action set.

5.5 Permission matrix

Capability	Patient	Clinician	Reviewer	Org	Payer	Demo Admin
Own person view	Yes	Assigned	Evidence only	No	No	Yes
Clinician queue	No	Assigned	Review subset	No	No	Yes
Safety replay	Own history	Assigned	Yes	No	No	Yes
Aggregate metrics	Own only	Assigned panel	Quality subset	Own org	Covered cohort	All demo
Cost model	No	No	No	Optional aggregate	Aggregate	All demo
Fairness audit	No	Read limited	Yes	Read aggregate	Read aggregate	Yes
Planning review	No	Comment	Approve/reject	Comment	Comment	Yes
Seed manifest	No	No	Read QA	No	No	Yes
Reset data	No	No	No	No	No	Yes
Credential config	No	No	No	No	No	Status only

Production admin, security admin, support and data analyst should become separate purpose-limited roles. Demo Admin is intentionally broad only inside the isolated fabricated environment.

SECTION 6

Implementation and release plan

Build in vertical slices. Each slice includes event generation, projection, API, page, audit and tests so the demo never relies on disconnected mock data.

DESIGN RULE	Action first. Meaning second. Evidence third. Raw records last.

6.1 Recommended coding waves

Wave	Build	Exit evidence
0 Inventory	Confirm current routes, roles, tenant model, event schemas and projection versions	Gap list and ADRs; no duplicate subsystem
1 Demo auth	Persona catalog, six role identities, session claims, role dropdown and role landing	Cross-role negative tests and environment isolation
2 Manifest	240-profile manifest, dictionaries, stable seeds, version registry and QA schema	Counts and balance checks pass
3 Generator	Six-month events, safety fixtures, corrections and reset job	Stable event and projection hashes
4 Role pro jections	Patient, clinician, reviewer, organization, payer and admin demo views	Same events produce correct minimum-necessary views
5 Metrics	Cohort service, metric dictionary, suppression, missingness and chart contracts	Metric fixtures match hand calculations
6 Planning rules	Versioned rules, signal state machine, lineage and review UI	No person-level actions; full audit
7 Governa nce	Fairness audit, model registry shell, monitoring and stop controls	Reviewers can block or retire output
8 Demo hardening	Scripted scenarios, clock advance, nightly reset, export labels and presenter checklist	Cold-start rehearsal passes twice

6.2 Test program

Suite	Minimum checks
Login	Role list, identity match, generic failures, expiry, revocation, role switch, prod isolation
Tenancy	Cross-tenant reads, guessed ids, list endpoints, exports, caches and background jobs
Manifest	240 total, 60 per region, 40 per age band, 30 per archetype, valid assignments
Generator	Repeatability, date bounds, event order, no orphans, corrections, late arrivals, partials
Safety	Every fixed rule, precedence, immediate persistence, pause, review and bounded re-entry
Projection	Role minimum necessary, evidence links, stale and failed states, rebuild equality
Metrics	Numerator, denominator, eligibility, missingness, censoring, intervals and versions
Suppression	Primary and complementary suppression; no tooltip, export or percentage leakage
Fairness	Group coverage, unknown values, intersections, proxy checks and disparate error review
Planning	Rule thresholds, repeat windows, blocked actions, transition policy and retirement
UI	Keyboard, focus, contrast, responsive layout, fabricated labels, empty/error/loading states
Operations	Reset, rollback, hash comparison, clock advance, secret rotation and audit failure

Use the 240 people as scenario fixtures and add lower-level unit tests around them. A test population is not a substitute for test assertions.

6.3 Release gates

Gate	Pass condition	Block release when
Truth	All pages and exports say fabricated demo data	Any screen can be mistaken for real data
Security	No demo identity works outside demo; no role escalation	Any cross-role or cross-tenant leak
Safety	Demo uses production-equivalent deterministic rules	Demo bypass or relaxed safety rule
Reproducibility	Reset reproduces expected events and hashes	Manual repair or nondeterministic output
Data quality	Counts, assignments, timestamps and references validate	Orphan, invalid or contradictory story
Analytics	Every value carries n/N, window, missingness and versions	Metric calculated only in client
Fairness	Suppression, unknown handling and review paths pass	Protected-group output can drive person care
Planning	All signals are aggregate, inspectable and reversible	Any automatic treatment or access decision
Operations	Nightly reset, backups, audit alerts and rehearsal pass	Failed reset or silent audit failure
Clinical review	Wording and allowed actions approved for demo	Outcome or causal overclaim

6.4 Acceptance stories

Audience	Acceptance story
Presenter	I can choose a role, authenticate, and reach a populated role home without setup during the meeting.
Patient	I see only my fabricated history, next safe action and evidence; I never see cohort rankings.
Clinician	I can identify who needs attention, why, freshness, owner and allowed next action from the first view.
Reviewer	I can replay a fixed gate from event through policy, projection and page, then record review.
Organization	I can see access, engagement, outcomes and capacity with definitions and suppressed small cells.
Payer	I can separate observed outcomes, pending evidence, utilization and modeled cost without person-level records.
Demo Admin	I can reset, advance the scripted clock, verify hashes and see every failed check.
Clinical lead	I can reject a planning signal and prove it cannot change patient care.
Fairness reviewer	I can inspect representation, missingness, disparities and proxy risks across predeclared groups.
Engineer	I can rebuild every projection and metric from the event ledger and match expected results.

6.5 Presenter scenario script

Minute	Role	Story
0-2	Patient	Login, daily action, progress, one missing check-in and evidence
2-5	Clinician	Queue, one safety review, one changing patient, measures and module history
5-7	Reviewer	Replay fixed gate, inspect policy version, response and correction
7-10	Organization	Regional access funnel, capacity, engagement and outcomes
10-13	Payer	Covered cohort, utilization, observed change and modeled cost separation
13-16	Demo Admin	Show 240-person manifest, dataset health, reset and projection verification
16-20	Planning	Open module signal, cohort, fairness review, limitations and pilot workflow

Keep two scripted paths: a 10-minute investor path and a 25-minute clinical and engineering path. Both start from a fresh reset. Do not rely on presenter memory to create the right data state.

6.6 Open decisions that do not block the first build

Decision	Default now	Later owner
Partner regions	Use Census regions	Product plus partner operations
Organization role	Include as sixth demo role	Product
Identity provider	Use isolated demo realm or tenant	Security
Demo credential distribution	Private presenter card and environment secrets	Security and sales operations
Outcome instruments	Use existing Steady measures and explicit versions	Clinical review
Responder threshold	Configuration fixture only; no clinical claim	Clinical and analytics
Internal subgroup minimum	n at least 30 by default	Privacy, statistics and fairness review
External suppression	Suppress 1 through 10 plus complementary cells	Privacy and payer contracting
Shadow models	Registry shell only in first release	Model governance group
Pilot method	Predeclared controlled rollout when enough data exists	Clinical, statistics and partner

6.7 Final coding rules

- Confirm existing code and schemas before adding any new subsystem.
- Keep demo data in isolated tenancy and visibly labeled on every surface.
- The role dropdown communicates intent; the server identity grants access.
- Use one fabricated event ledger and generate every role view from it.
- Make all random-looking data deterministic, seeded and versioned.
- Persist safety-relevant answers and evaluate fixed gates immediately.
- Never let AI, analytics or a planning rule clear or weaken a safety decision.
- Keep observed, adjusted, pilot and modeled results visually separate.
- Show numerator, denominator, time, cohort, missingness, suppression and versions.
- Use demographic data to test access and fairness, not to ration or select person-level care.
- Turn associations into reviewable hypotheses and controlled pilots, not treatment orders.
- Record every privileged read, export, reset, review and rule transition.
- Fail closed when identity, consent, tenant, purpose, provenance, definition or audit is missing.

The demo should feel real because its stories are coherent and traceable, not because it hides that the people and results are fabricated.

Sources and regulatory design anchors

These sources shape the governance and privacy posture. They do not by themselves establish that Steady is certified, validated, HIPAA-ready or approved for clinical decision support.

[1] U.S. Census Bureau, Geographic Levels and four Census Regions
https://www.census.gov/programs-surveys/economic-census/guidance-geographies/levels.html

[2] HHS, 2024 Section 1557 final rule, patient care decision support tools, 45 CFR 92.210
https://www.federalregister.gov/documents/2024/05/06/2024-08711/nondiscrimination-in-health-programs-and-activities

[3] ASTP/ONC, HTI-1 Final Rule and algorithm transparency
https://healthit.gov/regulations/hti-rules/hti-1-final-rule/

[4] ASTP/ONC, Decision Support Interventions and Predictive Models fact sheet
https://healthit.gov/resources/decision-support-interventions-dsi-fact-sheet-health-data-technology-and-interoperability-certification-program-updates-algorithm-transparency-and-information-sharing-hti-1-final-rule/

[5] NIST, AI Risk Management Framework 1.0
https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf

[6] NIST, Towards a Standard for Identifying and Managing Bias in AI
https://nvlpubs.nist.gov/NISTpubs/SpecialPublications/NIST.SP.1270.pdf

[7] AHRQ, Guiding Principles for health care algorithms and racial or ethnic bias
https://www.ahrq.gov/news/newsroom/press-releases/guiding-principles.html

[8] HHS OCR, HIPAA de-identification guidance
https://www.hhs.gov/hipaa/for-professionals/special-topics/de-identification/index.html

[9] CMS, small-cell suppression example for values below 11
https://data.cms.gov/resources/managing-clinician-aggregation-group-performance-data-dictionary